BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 138 · Lesson 138 · SQL SELECT, Filtering, Sorting and JOIN

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What table is created to remove an EmployeeID→DeptID→DeptName transitive dependency?

Q2. When does a linear search stop?

Q3. Which is removed first from a FIFO queue?

Q4. What does ROM stand for?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

SELECT — Returns specified columns/expressions.

WHERE — Filters rows according to a condition.

ORDER BY — Sorts returned rows by chosen column/expression.

JOIN — Combines rows from related tables using a join condition.

New learning

Choose projection — SELECT chooses columns from a named relation.

Filter and order — WHERE restricts rows; ORDER BY arranges results.

Join related rows — Link keys using INNER JOIN ... ON.

Worked example

Orders(OrderID,CustomerID,Total) has totals 25,75,120. Write SQL listing orders over 50 with largest first.

1. Filter Total > 50 — Find the given data and name the required result.

2. Select OrderID, Total — Explain why this intermediate stage is necessary.

3. Sort DESC 120,75 — SELECT OrderID, Total FROM Orders WHERE Total > 50 ORDER BY Total DESC; returns orders with 120 then 75, excluding 25.

We Do · Practise with guidance

Customer(CustomerID,Name), Orders(OrderID,CustomerID,Total). Show customer names for orders over 100.

Join on CustomerID

Find the given data and name the required result.

Filter Total > 100

Explain why this intermediate stage is necessary.

Project Customer.Name

SELECT Customer.Name FROM Customer INNER JOIN Orders ON Customer.CustomerID = Orders.CustomerID WHERE Orders.Total > 100; Returns a row per matching order; DISTINCT can remove repeated customer names.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Write SQL listing Title and Price from Book where Price<10, sorted ascending by Price. [6]

Question 2

Explain the role of the ON clause in an INNER JOIN between Borrower and Loan. [6]

Question 3

Write an INNER JOIN query listing Student.Name and Course.Title using Student, Enrolment and Course. [8]

Exit Ticket · Answer before checking

ORDER BY always removes duplicate result rows.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Department(DeptID,DeptName), referenced by Employee.

Q2. On a match or after examining all items.

Q3. The longest-waiting item at the front.

Q4. Read-only memory.

You Do pages · Independent questions

Q1. SELECT Title,Price [2] FROM Book [1] WHERE Price <10 [2] ORDER BY Price ASC [1].

Q2. Join matches records [2], ON states matching FK/PK relationship [2], avoids Cartesian result/mismatched associations [2].

Q3. SELECT columns [2], FROM Student JOIN Enrolment ON Student.StudentID=Enrolment.StudentID [3], JOIN Course ON Enrolment.CourseID=Course.CourseID [3].

Exit Ticket · Model answer

False. Sorting changes order, not duplication; DISTINCT may remove duplicates.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.